MATH 330: Measure-Theoretic Probability Theory
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Chapter 1

Introduction and Motivation

Definition 1.1. A probability space is a triple $(\Omega, \mathcal{F}, P)$ where Ω is the sample space, $\mathcal{F}$ is a collection of sets in which we can assign a probability and P is the probability function. We will expand this definition in the next chapter.

Remark 1.1. In a finite sample space we have $\mathcal{F} = 2^{\Omega}$.

Example 1.1. Consider a random chord of length ℓ in a circle. Find $P[\ell > \sqrt{3}]$. We will discuss three different methods to solve this.

(1) Method 1: Inscribe an equilateral triangle in the circle. Take the perpendicular from the vertex of the triangle. Then the probability becomes

$$P[\ell > \sqrt{3}] = \frac{60}{180} = \frac{1}{3}$$

. Where 60 is the angle of the triangle and 180 is the total angle of the circle.

(2) Method 2: Inscribe an equilateral triangle in the circle and the midpoint of its height. Then the probability becomes

$$P[\ell > \sqrt{3}] = \frac{1}{2}$$

.

(3) Method 3: Inscribe an equilateral triangle in the circle and inscribe a circle inside the triangle then the probability becomes

$$P[\ell > \sqrt{3}] = \frac{\text{area of small circle}}{\text{area of big circle}} = \frac{1}{4}$$

.

Example 1.2. What is the probability of choosing the number 7 from a uniform distribution over $\mathbb{N}$? The answer is 0

- if $p > 0$ then for large n $np > 1$ which is absurd.
- if $p = 0$ then $\sum_{n=1}^{\infty} P(\{n\}) = 0$

Example 1.3. Let $\Omega = [0, 1]$ and choose X uniformly from Ω. Then $P[X = x] = 0 \quad \forall x \in \Omega$, but $P[X \in [0, \frac{1}{2}]] = \frac{1}{2}$. Why? because its continuous. We can't assign probability to intervals from points because an interval is an uncountable collection of points. So we need to assign probability to the interval itself.

Remark 1.2. We can constract a set for which we can't assign a probability. For example, consider the set $A = \{x + y\sqrt{2} : x, y \in \mathbb{Q}\}$. Then A is dense in $\mathbb{R}$ and we can't assign a probability to it.

1.1 Expectation

Definition 1.2. The discrete expectation of a random variable X is defined as

$$\mathbb{E}[X] = \sum_{x \in \Omega} xP(X = x)$$

Definition 1.3. The absolute continuous expectation of a random variable X is defined as

$$\mathbb{E}[X] = \int_{-\infty}^{\infty} x f_X(x) dx$$

where $f_X(x)$ is the probability density function of X.

Remark 1.3. In both cases this only holds if $\int_{-\infty}^{\infty} |x| f_X(x) dx < \infty$ absolute convergence.

1.2 Random variable

Definition 1.4. A random variable is a function $X : \Omega \to \mathbb{R}$ that assigns a real number to each outcome in the sample space Ω.

$$X : \Omega \to \mathbb{R} \qquad \omega \mapsto X(\omega)$$

and the expected value is

$$\mathbb{E}[X] = \int_{\Omega} X(\omega) dP(\omega) = \int_{\Omega} X(\omega) P(d\omega)$$

Chapter 2

Measure theory

Definition 2.1. A field or algebra is a collection of sets $\mathcal{C}$ satisying the following:

(1) $\Omega \in \mathcal{C}$

(2) If $A \in \mathcal{C}$ then $A^c \in \mathcal{C}$

(3) If $A, B \in \mathcal{C}$ then $A \cup B \in \mathcal{C}$

Definition 2.2. A $\sigma -$ field or a $\sigma-$algebra is a collection of sets $\mathcal{F}$ satisfying the following:

(1) $\Omega \in \mathcal{F}$

(2) If $A \in \mathcal{F}$ then $A^c \in \mathcal{F}$

(3) If $A_1, A_2, \ldots \in \mathcal{F}$ then $\bigcup_{i=1}^{\infty} A_i \in \mathcal{F}$

Remark 2.1. Some are the following properties of fields and $\sigma-$algebras which are simple to prove:

(1) $A_1, A_2, \ldots, A_n \in \mathcal{C} \implies \bigcup_{i=1}^{n} A_i \in \mathcal{C}$

(2) $A_1, A_2, \ldots, A_n \in \mathcal{C} \implies \bigcap_{i=1}^{n} A_i \in \mathcal{C}$

(3) $A_1, A_2, \ldots, A_n \in \mathcal{F} \implies \bigcap_{i=1}^{\infty} A_i \in \mathcal{F}$

Remark 2.2. If $\mathcal{F}$ is a $\sigma-$algebra then $\mathcal{F}$ is a field. But the converse is not true. For example, consider $\Omega = \mathbb{R}$ and $\mathcal{C} = \{A \subseteq \mathbb{R} : A \text{ is finite or } A^c \text{ is finite}\}$. Then $\mathcal{C}$ is a field but not a $\sigma-$algebra.

Definition 2.3. Given a collection C of sets then the $\sigma -$ field generated by C is denoted by $\sigma(C)$ and in fact is the smallest $\sigma -$ field that contains C. In other words, if $\mathcal{F}$ is a $\sigma -$ field such that $C \subseteq \mathcal{F}$ then $\sigma(C) \subseteq \mathcal{F}$.

Proof. Let $\aleph = \{\mathcal{F} : \mathcal{F}\text{is a } \sigma - \text{field and } \mathcal{F} \supseteq C\}$. Then $\aleph$ is nonempty because $2^{\mathcal{C}}$ is a $\sigma -$ field that contains $\mathcal{C}$. Define $\sigma(\mathcal{C}) = \bigcap_{\mathcal{F} \in \aleph} \mathcal{F}$ □

Note 2.1. 2^{Ω} is a $\sigma -$ field

Remark 2.3. The following are some remarks on $\sigma -$ fields

(1) If F_1, F_2 are $\sigma -$ fields then $F_1 \cap F_2$ is a $\sigma -$ field.

(2) If F_1, F_2 are $\sigma -$ fields then $F_1 \cup F_2$ is not necessarily a $\sigma -$ field.

(3) If F_α is a collection of $\sigma -$ fields then $\bigcap_\alpha F_\alpha$ is a $\sigma -$ field.

Example 2.1. Let $\Omega = \mathbb{R}$, $A = [0,1]$, $B = [3,4]$ then $\sigma(\{A\}) = \{\Omega, \emptyset, A, A^c\}$ and $\sigma(\{B\}) = \{\Omega, \emptyset, B, B^c\}$. But $\sigma(\{A\}) \cup \sigma(\{B\}) = \{\Omega, \emptyset, A, A^c, B, B^c\}$ is not a $\sigma -$ field.

Definition 2.4. The Borel $\sigma -$ field-algebra is the $\sigma -$ field-algebra generated by the open sets in $\mathbb{R}$. We denote it by $\mathcal{B}(\mathbb{R})$.

Remark 2.4. The Borel $\sigma -$ field can be generated by the closed sets, the open intervals, the closed intervals, the half-open intervals, and the half-closed intervals.

2.1 Limit of sets

Definition 2.5. A sequence of sets $\{A_n\}_{n=1}^{\infty}$ with $A_n \subset \Omega$. We define the following:

(1) $\inf_{k\geq n} A_k = \bigcap_{k\geq n} A_k$

(2) $\sup_{k\geq n} A_k = \bigcup_{k\geq n} A_k$

(3) $\liminf_{n\to\infty} A_n = \sup_{n\geq 1} \inf_{k\geq n} A_k = \bigcup_{n=1}^{\infty} \bigcap_{k\geq n} A_k$

(4) $\limsup_{n\to\infty} A_n = \inf_{n\geq 1} \sup_{k\geq n} A_k = \bigcap_{n=1}^{\infty} \bigcup_{k\geq n} A_k$

Remark 2.5. We can notice the following :

(1) $\bigcap_{k=1}^{\infty} A_k, \bigcup_{k=2}^{\infty} A_k, \ldots$ is increasing

(2) $\bigcup_{k=1}^{\infty} A_k, \bigcap_{k=2}^{\infty} A_k, \ldots$ is decreasing

(3) All the above are sets in the $\sigma -$ field if $A, B \in \mathcal{B}$. (Borel $\sigma -$ field)

(4) $(\liminf_{n\to\infty} A_n)^c = (\bigcap_{n=1}^{\infty} \bigcup_{k\geq n} A_k^c) = \limsup_{n\to\infty} A_n^c$

Definition 2.6. The limit of $A_1, A_2, \ldots$ exists if and only if $\liminf_{n\to\infty} A_n = \limsup_{n\to\infty} A_n$ and we denote it by $\lim_{n\to\infty} A_n$.

Example 2.2.

$$\liminf_{n\to\infty} \left[0, \frac{n}{n+1}\right) = \bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} \left[0, \frac{k}{k+1}\right) = \bigcup_{n=1}^{\infty} \left[0, \frac{n}{n+1}\right) = [0,1),$$

$$\limsup_{n\to\infty} \left[0, \frac{n}{n+1}\right) = \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} \left[0, \frac{k}{k+1}\right) = \bigcap_{n=1}^{\infty} [0,1) = [0,1).$$

Therefore, $\lim_{n\to\infty} \left[0, \frac{n}{n+1}\right) = [0,1)$.

Lemma 2.1.

$$\begin{aligned}\limsup_{n\to\infty} A_n &= \left\{\omega \in \Omega : \sum_{n=1}^{\infty} \mathbb{1}_{A_n}(\omega) = \infty\right\} \\ &= \{\omega : \omega \in A_{n_k} \textit{ for infinitely many } k\} = \{A_n \textit{ i.o.}\},\end{aligned}$$

where (n_k) is a subsequence depending on ω and $\mathbb{1}$ is the indicator function.

Proof. Suppose that $\omega \in \limsup_{n\to\infty} A_n = \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k$. Then for all n we have $\omega = \bigcup_{k=n} A_k$ □

Remark 2.6. $\liminf_{n\to\infty} A_n \subseteq \limsup_{n\to\infty} A_n$

2.1.1 Monotone sequence of sets

Definition 2.7. We say that A_n is increasing if $A_i \subset A_{i+1}$ for all $i \in \mathbb{N}$, and we say that A_n is decreasing if $A_i \supset A_{i+1}$ for all $i \in \mathbb{N}$.

Proposition 2.1. *We define the following:*

(1) If A_n is increasing, then $\lim_{n\to\infty} A_n = A$ and $A = \bigcup_{n=1}^{\infty} A_n$.

(2) If A_n is decreasing, then $\lim_{n\to\infty} A_n = A$ and $A = \bigcap_{n=1}^{\infty} A_n$.

Proof. We prove the first part of the above proposition:

(1) For an increasing sequence,

$$\liminf_{n\to\infty} A_n = \bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} A_k = \bigcup_{n=1}^{\infty} A_n,$$
$$\limsup_{n\to\infty} A_n = \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k = \bigcup_{n=1}^{\infty} A_n.$$

Indeed, if an element belongs to some A_m, then it belongs to every A_k with $k \geq m$. Therefore, $\limsup A_n = \liminf A_n = \lim A_n = \bigcup_{n=1}^{\infty} A_n$.

□

Corollary 2.1. *For any sequence of sets A_n, we have*

$$\liminf A_n = \bigcup_{n=1}^{\infty} \bigcap_{k=n}^{\infty} A_k = \lim_{n\to\infty} (\inf_{k\geq n} A_k)$$

Similarly,

$$\limsup A_n = \bigcap_{n=1}^{\infty} \bigcup_{k=n}^{\infty} A_k = \lim_{n\to\infty} (\sup_{k\geq n} A_k)$$

Remark 2.7. The inclusion-exclusion principle is a fundamental tool in probability theory. It allows us to compute the probability of the union of events by considering the probabilities of their intersections. To show that a sequence of random variables X_n converges to X_0 almost surely (a.s), i.e

$$P[\{\omega : X_n(\omega) \to X_0(\omega)\}] = 1$$

Then we need equivalently to prove that $\forall \varepsilon > 0$ we have

$$P[|X_n(\omega) - X_0(\omega) > \varepsilon \quad \text{i.o}] = 0$$

2.2 Probability Space

Definition 2.8. A probability sspace is a triple $(\Omega, \mathcal{F}, P)$ where Ω is the sample space, $\mathcal{B}$ is a σ-algebra of events, and P is a probability measure.

Axioms 2.1. $P : \mathcal{B} \to \mathbb{R}$ such that

(1) $P(A) \geq 0$ for all $A \in \mathcal{B}$

(2) $P(\Omega) = 1$

(3) $\sigma-$additivity: If $\{A_j\}_{j=1}^{\infty}$ is a sequence of disjoint sets in $\mathcal{B}$, then $P\left(\bigcup_{j=1}^{\infty} A_j\right) = \sum_{j=1}^{\infty} P(A_j)$.

Theorem 2.1 (Basic Theorems)**.** *The following are some of the basic theorems:*

(1) $P(\emptyset) = 0$

Proof. Let $A_1 = \Omega$ and $A_n = \emptyset$ for all $n \geq 2$. Then by $\sigma-$additivity we have

$$P(\Omega) = P\left(\bigcup_{n=1}^{\infty} A_n\right) = \sum_{n=1}^{\infty} P(A_n) = P(\Omega) + \sum_{n=2}^{\infty} P(\emptyset)$$

which implies that $P(\emptyset) = 0$. □

(2) Finite additivity: If $A_1, A_2, \ldots, A_n$ are disjoint sets in $\mathcal{B}$, then $P\left(\bigcup_{j=1}^{n} A_j\right) = \sum_{j=1}^{n} P(A_j)$.

Proof. Let $A_1, A_2, \ldots, A_n$ be disjoint sets in $\mathcal{B}$. Define $A_{n+1} = A_{n+2} = \ldots = \emptyset$. Then by $\sigma-$additivity we have

$$P\left(\bigcup_{j=1}^{\infty} A_j\right) = \sum_{j=1}^{\infty} P(A_j) = \sum_{j=1}^{n} P(A_j) + \sum_{j=n+1}^{\infty} P(\emptyset) = \sum_{j=1}^{n} P(A_j)$$

which implies that $P\left(\bigcup_{j=1}^{n} A_j\right) = \sum_{j=1}^{n} P(A_j)$. □

(3) Inclusion - Exclusion: For 2 events:

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Proof. $P(A_1) = P(A_1A_2) + P(A_1A_2^c)$ and $P(A_2) = P(A_1A_2) + P(A_1^cA_2)$ then

$$P(A_1) + P(A_2) = P(A_1A_2) + P(A_1A_2^c) + P(A_1A_2) + P(A_1^cA_2) = P(A_1A_2) + P(A_1 \cup A_2)$$

which implies that

$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

Then by induction we can prove the general case for n events which is

$$P\left(\bigcup_{i=1}^{n} A_i\right) = \sum_{k=1}^{n} \sum_{1 \leq i_i < \ldots < i_k \leq n} (-1)^{k+1} P\left(\bigcap_{j=1}^{k} A_{i_j}\right)$$

□

(4) Bonferonni's Inequalities:

(1)

$$P(\bigcup_{i=1}^{n} A_i) \leq \sum_{i=1}^{n} P(A_i)$$

(2)

$$P(\bigcap_{i=1}^{n} A_i) \geq \sum P(A_i) - \sum_{1\leq i<j\leq n} P(A_i \cap A_j)$$

(5) Monotonicity:

$$P(\bigcup_{i=1}^{\infty} A_n) \leq \sum_{i=1}^{\infty} P(A_i)$$

Proof. Let $B_1 = A_1$ and $B_i = A_i \setminus \bigcup_{j=1}^{i-1} A_j$ for $i \geq 2$. Then $\bigcup_{i=1}^{n} A_i = \bigcup_{i=1}^{n} B_i$ and the sets B_i are disjoint. Therefore, by finite additivity,

$$P\left(\bigcup_{i=1}^{n} A_i\right) = P\left(\bigcup_{i=1}^{n} B_i\right) = \sum_{i=1}^{n} P(B_i) \leq \sum_{i=1}^{n} P(A_i)$$

which implies that $P(\bigcup_{i=1}^{\infty} A_n) \leq \sum_{i=1}^{\infty} P(A_i)$. □

(6) Continuity of probability for monotone sequences: Let A_i be increasing and $A_i \in B$ then $P(\lim A_i) = \lim P(A_i)$

Proof. — Homework □

(7) Fatou's Lemma: Let $A_n \in B$ then:

$$P(\liminf A_n) \leq \liminf P(A_n) \leq \limsup P(A_n) \leq P(\limsup A_n)$$

Idea of proof. Since $\liminf A_n = \bigcup_{n=1}^{\infty} \bigcap_{k\geq n} A_k$, by continuity from below,

$$P(\liminf A_n) = \lim_{n\to\infty} P\left(\bigcap_{k\geq n} A_k\right).$$

As $\bigcap_{k\geq n} A_k \subseteq A_n$, we have $P\left(\bigcap_{k\geq n} A_k\right) \leq P(A_n)$ for each n, which gives us

$$P(\liminf A_n) \leq \liminf_{n\to\infty} P(A_n).$$

Similarly, since $\limsup A_n = \bigcap_{n=1}^{\infty} \bigcup_{k\geq n} A_k$, by continuity from above,

$$P(\limsup A_n) = \lim_{n\to\infty} P\left(\bigcup_{k\geq n} A_k\right).$$

As $A_n \subseteq \bigcup_{k\geq n} A_k$, we have $P(A_n) \leq P\left(\bigcup_{k\geq n} A_k\right)$ for each n, hence we can prove that

$$\limsup_{n\to\infty} P(A_n) \leq P(\limsup A_n).$$

□

Remark 2.8. $A_n \uparrow$ or $A_n \downarrow$ is sufficient for the limit to exist.

Definition 2.9. Let $\Omega = \mathbb{R}$ and P be a probability measure on $\mathbb{R}$ and define $F : \mathbb{R} \to \mathbb{R}$ such that $F(x) = P((-\infty, x])$. Then, F has the following properties:

(1) F is non-decreasing in $\mathbb{R}$

(2) F is right-continuous

(3) $\lim_{x\to-\infty} F(x) = 0$ and $\lim_{x\to\infty} F(x) = 1$

Proof. (1) To show that F is non-decreasing, let $x < y$. Then $(-\infty, x] \subseteq (-\infty, y]$, so $F(x) = P((-\infty, x]) \leq P((-\infty, y]) = F(y)$.

(2) To show that F is right-continuous, let $x_n \downarrow x$. Then $(-\infty, x_n] \downarrow (-\infty, x]$, so by continuity from above, $F(x_n) \to F(x)$.

(3) As $x \to -\infty$, we have $(-\infty, x] \to \emptyset$, so $F(x) \to 0$. As $x \to \infty$, we have $(-\infty, x] \to \mathbb{R}$, so $F(x) \to 1$.

□

2.2.1 Other structures

Definition 2.10. Consider a class $\mathcal{S}$ of subsets of Ω. We call $\mathcal{S}$ a π-system if it is closed under finite intersections, i.e. if $A, B \in \mathcal{S}$ then $A \cap B \in \mathcal{S}$ and by induction $A_1 \cap A_2 \cap \cdots \cap A_n \in \mathcal{S}$ for any finite collection of sets in $\mathcal{S}$. We call $\mathcal{S}$ a λ-system if it satisfies the following:

(1) $\Omega \in \mathcal{S}$

(2) If $A, B \in \mathcal{S}$ and $A \subseteq B$ then $B \setminus A \in \mathcal{S}$

(3) $A_1 \subseteq A_2 \subseteq \cdots$ and $A_i \in \mathcal{S}$ for all i then $\lim \uparrow A_i = \bigcup_{i=1}^{\infty} A_i \in \mathcal{S}$

Equivalently we can define a λ-system as follows:

(1) $\Omega \in \mathcal{S}$

(2) If $A \in \mathcal{S}$ then $A^c \in \mathcal{S}$

(3) If $A_1, A_2, \ldots$ is a sequence of disjoint sets in $\mathcal{S}$ then $\bigcup_{n=1}^{\infty} A_n \in \mathcal{S}$

Proof of the equivalence of the definitions. Suppose that $\mathcal{S}$ is a λ-system according to the first definition. Then $\Omega \in \mathcal{S}$ and if $A \in \mathcal{S}$ then $A^c = \Omega \setminus A \in \mathcal{S}$. If $A_1, A_2, \ldots$ is a sequence of disjoint sets in $\mathcal{S}$ then we can define $B_n = \bigcup_{i=1}^{n} A_i$. Then $B_n \subseteq B_{n+1}$ and $B_n \in \mathcal{S}$ for all n. Therefore, $\lim_{n\to\infty} B_n = \bigcup_{n=1}^{\infty} A_n \in \mathcal{S}$.

Conversely, suppose that $\mathcal{S}$ is a λ-system according to the second definition. Then $\Omega \in \mathcal{S}$ and if $A, B \in \mathcal{S}$ and $A \subseteq B$ then $B \setminus A = B \cap A^c \in \mathcal{S}$. If $A_1, A_2, \ldots$ is a sequence of sets in $\mathcal{S}$ such that $A_1 \subseteq A_2 \subseteq \cdots$, then we can define $B_n = A_n$ for all n. Then B_n is a sequence of disjoint sets in $\mathcal{S}$ and $\lim_{n\to\infty} B_n = \bigcup_{n=1}^{\infty} A_n \in \mathcal{S}$. □

Definition 2.11. Given a class $\mathcal{C}$ and a structure $\mathcal{S}$ being one of the structures defined previously, then $\mathcal{S}(\mathcal{C})$ is the smallest structure that contains $\mathcal{C}$, i.e. if $\mathcal{S}'$ is a structure such that $\mathcal{C} \subseteq \mathcal{S}'$ then $\mathcal{C} \subseteq \mathcal{S}(\mathcal{C})$ and $\mathcal{S}(\mathcal{C}) \subseteq \mathcal{S}'$.

Proposition 2.2. *Let P_1, P_2 be two probability measures on $(\Omega, \mathcal{B})$ and define*

$$\mathcal{L} = \{A : P_1(A) = P_2(A)\}$$

then $\mathcal{L}$ is a λ-system.

Proof. We need to show that $\mathcal{L}$ satisfies the three properties of a λ-system.

(1) $\Omega \in \mathcal{L}$ because $P_1(\Omega) = P_2(\Omega) = 1$.

(2) If $A \in \mathcal{L}$ then $P_1(A) = P_2(A)$ and $P_1(A^c) = 1 - P_1(A) = 1 - P_2(A) = P_2(A^c)$, so $A^c \in \mathcal{L}$.

(3) If $A_1, A_2, \ldots$ is a sequence of disjoint sets in $\mathcal{L}$, then $P_1(A_i) = P_2(A_i)$ for all i. By σ-additivity, we have
$$P_1\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P_1(A_i) = \sum_{i=1}^{\infty} P_2(A_i) = P_2\left(\bigcup_{i=1}^{\infty} A_i\right)$$
so $\bigcup_{i=1}^{\infty} A_i \in \mathcal{L}$.

□

Theorem 2.2 (Dynkin's π-λ theorem)**.** *If $\mathcal{P}$ is a π-system and $\mathcal{L}$ is a λ-system such that $\mathcal{P} \subseteq \mathcal{L}$, then $\sigma(\mathcal{P}) \subseteq \mathcal{L}$. Moreover, if $\mathcal{C}$ is a π-system, then $\sigma(\mathcal{C}) = \lambda(\mathcal{C})$.*

Corollary 2.2. *If P_1, P_2 are probability measures on $(\Omega, \mathcal{B})$ and if $\mathcal{P}$ is a π-system such that $P_1(A) = P_2(A)$ for all $A \in \mathcal{P}$, then $P_1(B) = P_2(B)$ for all $B \in \sigma(\mathcal{P})$.*

Proof. Follows from the previous proposition and Dynkin's π-λ theorem. □

Corollary 2.3. *$\mathcal{P} = \{(-\infty, x] : x \in \mathbb{R}\}$ is a π-system and for all x, y we have*
$$(-\infty, x] \cap (-\infty, y] = (-\infty, x \wedge y].$$
Also P generates the Borel sets. If F_1, F_2 are two cumulative distribution functions that agree on each x then $F_1(x) = F_2(x)$ for all x and therefore $P_1((\infty, x]) = P_2((-\infty, x])$ where $P_i(-\infty, x]) = F_i(x)$ for $i = 1, 2$.

Proof of Dynkin's π-λ theorem.
Claim (Claim 0). It is enough to show that $\lambda(\mathcal{P})$ is a π-system.

Proof of Claim 0. If $\lambda(\mathcal{P})$ is a π-system, since its also a λ-system, then it is a σ-algebra. Since $\mathcal{P} \subseteq \lambda(\mathcal{P})$ and $\lambda(\mathcal{P})$ is now a σ-algebra, by minimality of $\sigma(\mathcal{P})$ we have $\sigma(\mathcal{P}) \subseteq \lambda(\mathcal{P})$. Now, since $\mathcal{P} \subseteq \mathcal{L}$ and $\mathcal{L}$ is a λ-system, by minimality of $\lambda(\mathcal{P})$ we have $\lambda(\mathcal{P}) \subseteq \mathcal{L}$, and therefore $\sigma(\mathcal{P}) \subseteq \mathcal{L}$. □

We focus on proving that $\lambda(\mathcal{P})$ is a π-system. Let $A \in \mathcal{B}$ and define
$$\mathcal{G}_A = \{B \in \mathcal{B} : A \cap B \in \lambda(\mathcal{P})\}$$

Claim (Claim 1). If $A \in \lambda(\mathcal{P})$ then $\mathcal{G}_A$ is a λ-system.

Proof of Claim 1. We need to check that:

- $\Omega \in \mathcal{G}_A$ because $A \cap \Omega = A \in \lambda(\mathcal{P})$.
- Suppose $B \in \mathcal{G}_A$, so $A \cap B \in \lambda(\mathcal{P})$. Since $A \cap B \subseteq A$ and both $A \cap B, A \in \lambda(\mathcal{P})$, and $\lambda(\mathcal{P})$ is a λ-system (closed under differences of nested sets), we get $A \cap B^c = A \setminus (A \cap B) \in \lambda(\mathcal{P})$, i.e. $B^c \in \mathcal{G}_A$.
- Let $\{B_j\}$ $j \geq 1$ be disjoint such that $B_j \in \mathcal{G}_A$ for all j. Then, $A \cap B_j \in \lambda(\mathcal{P})$ for all j and $A \cap B_j$ are disjoint. Since $\lambda(\mathcal{P})$ is a λ-system, we have $\bigcup_{j=1}^{\infty}(A \cap B_j) = A \cap \left(\bigcup_{j=1}^{\infty} B_j\right) \in \lambda(\mathcal{P})$, so $\bigcup_{j=1}^{\infty} B_j \in \mathcal{G}_A$.

□

Claim (Claim 2). If $A \in \mathcal{P}$ then $\mathcal{G}_A$ contains $\mathcal{P}$.

Proof. First, observe that $A \in \mathcal{P} \subseteq \lambda(\mathcal{P})$, but by Claim 1, $\mathcal{G}_A$ is a λ-system. Now, for any $B \in \mathcal{P}$, since $\mathcal{P}$ is a π-system, we have $A \cap B \in \mathcal{P} \subseteq \lambda(\mathcal{P})$. Therefore, $B \in \mathcal{G}_A$, which shows that $\mathcal{G}_A$ contains all of $\mathcal{P}$. □

By minimality of $\lambda(\mathcal{P})$, we have $\lambda(\mathcal{P}) \subseteq \mathcal{G}_A$. In particular, since $A \in \lambda(\mathcal{P})$, we have $A \in \mathcal{G}_A$, which means that $A \cap A = A \in \lambda(\mathcal{P})$. Now, for any $B \in \lambda(\mathcal{P})$, we have $B \in \mathcal{G}_A$ by the previous argument, which means that $A \cap B \in \lambda(\mathcal{P})$. This shows that $\lambda(\mathcal{P})$ is closed under finite intersections, and hence is a π-system.

□

2.3 Construction of Probability Measures

Skeleton 2.1. Given a probability measure P on a semi-algebra $\mathcal{S}$ we can extend to a probability measure P_1 on the generated algebra $\mathcal{F} = \mathcal{A}(\mathcal{S})$. We have that $P_1 \mid \mathcal{S} = P$, we can extend once again to a probability measure on $\sigma(\mathcal{F})$.

Definition 2.12. A semi-algebra $\mathcal{S}$ is a class of subsets of Ω such that:

(1) $\emptyset, \Omega \in \mathcal{S}$

(2) $\mathcal{S}$ is a π-system, i.e., if $A, B \in \mathcal{S}$ then $A \cap B \in \mathcal{S}$

(3) If $A \in \mathcal{S}$ then A^c can be expressed as a finite disjoint union of sets in $\mathcal{S}$, i.e. $A^c = \bigcup_{i=1}^n C_i$ where $C_i \in \mathcal{S}$ and $C_i \cap C_j = \emptyset$ for $i \neq j$.

Remark 2.9. This structure is motivated by the intervals $(a, b]$ where $-\infty \leq a \leq b \leq \infty$. Note that $(-\infty, b] \in \mathcal{S}$ where $\mathcal{S} = \{(a, b] : -\infty \leq a \leq b \leq \infty\}$, $(a, b]^c = (-\infty, a] \cup (b, \infty)$ obviously disjoint.

Definition 2.13. A rectangle A in $\mathbb{R}^k$ has the form

$$A = I_1 \times I_2 \times \cdots \times I_k$$

where I_i is an interval of the form $(a, b]$. We write $I_j \in S_1$ where

$$S_1 = \{(a, b] : -\infty \leq a \leq b \leq \infty\}$$

. Obviously, $\emptyset, \Omega$ belong to the collection of rectangles

$$S_k = \{A = I_1 \times I_2 \times \cdots \times I_k : I_i \in S_1\}$$

and it is clear that the intersection of rectangles is a rectangle. In general,

$$A = I_1 \times I_2 \times \cdots \times I_k = \{(x_1, \cdots, x_k) : x_j \in I_j \quad \forall j = 1, \ldots, k\}$$

and

$$A^c = \left(\bigcap_{j=1}^{k} \{(x_1, \ldots, x_n) : x_j \in I_j\} \right)^c = \bigcup_{j=1}^{k} \{(x_1, \ldots, x_n) : x_j \in I_j^c\}$$

. Since each of the sets $\{x_1, \ldots, x_k) : x_j \in I_j^c\}$ are disjoint rectangles, this proves the 3rd property of a semi-algebra.

Lemma 2.2. *Given a semi-algebra $\mathcal{S}$, let $\mathcal{A}(S)$ be the smallest algebra generated by $\mathcal{S}$. Define*

$$\mathcal{A}(\mathcal{S}) = \{\sum_{i\in I} S_i : I \text{ is a finite index set and } S_i \in \mathcal{S}\}$$

where $\sum_{i\in I} S_i$ is a disjoint union of sets in $\mathcal{S}$ and denote

$$\Lambda = \{\sum_{i\in I} S_i : I \text{ is a finite index set and } S_i \in \mathcal{S}\}$$

Proof. It is clear that each $A \in \Lambda$ is a union of sets in $\mathcal{S}$ so a union of sets in $\mathcal{A}(\mathcal{S})$ hence $A \in \mathcal{A}(\mathcal{S})$ by the additivity of $\mathcal{A}(\mathcal{S})$ and the finiteness of the index set I, thus $\Lambda \subseteq \mathcal{A}(\mathcal{S})$. Now we need to prove the other containement, clearly: each set $S \in \mathcal{S}$ belongs to Λ, take I to be a sigleton. Now, $\Omega \in \Lambda$ because $\Omega \in \mathcal{S}$. Now lets prove the 3rd property, if $\sum_{i\in I} S_i$ and $\sum_{j\in J} S_j$ are both in Λ. Then,

$$\left(\sum_{i\in I} S_i\right) \cap \left(\sum_{j\in J} S_j\right) = \sum_{(i,j)\in I\times J} S_i S_j$$

Since S_i and S_j are in $\mathcal{S}$, then $S_i \cap S_j \in \mathcal{S}$, and the index set $I \times J$ is finite, thus $\left(\sum_{i\in I} S_i\right) \cap \left(\sum_{j\in J} S_j\right) \in \Lambda$. Now, if $\sum_{i\in I} S_i \in \Lambda$, then $\left(\sum_{i\in I} S_i\right)^c = \bigcap_{i\in I} S_i^c$. Since S_i^c can be expressed as a finite disjoint union of sets in $\mathcal{S}$, we have that $\left(\sum_{i\in I} S_i\right)^c$ can also be expressed as a finite disjoint union of sets in $\mathcal{S}$, thus $\left(\sum_{i\in I} S_i\right)^c \in \Lambda$. Therefore, Λ is an algebra and since it contains $\mathcal{S}$, we have that $\mathcal{A}(\mathcal{S}) = \Lambda$. □

Theorem 2.3 (First Extension Theorem)**.** *Let $\mathcal{S}$ be a semi-algebra and let P be a probability measure on $\mathcal{S}$*

$$P : \mathcal{S} \to [0,1] \quad P(\Omega) = 1 \quad P \text{ is } \sigma - \text{additive}$$

Then P has a unique extension to a σ-additive probability measure on the algebra $\mathcal{A}(\mathcal{S})$ generated by $\mathcal{S}$.

Proof. Let $A \in \mathcal{A}(\mathcal{S})$. Then

$$A = \sum_{i\in I} S_i,$$

where I is a finite index set and $S_i \in \mathcal{S}$ for all i. Define

$$P'(A) = \sum_{i\in I} P(S_i).$$

P' is well-defined. Suppose that

$$A = \sum_{i\in I} S_i = \sum_{j\in J} S_j.$$

We need to show that

$$\sum_{i\in I} P(S_i) = \sum_{j\in J} P(S_j).$$

Since $S_i \subseteq A$, we have

$$S_i = S_i \cap A = S_i \cap \bigcup_{j\in J} S_j = \sum_{j\in J} S_i S_j.$$

So,

$$\sum_{i\in I} P(S_i) = \sum_{i\in I} P\left(\sum_{j\in J} S_i S_j\right).$$

By the additivity of P on $\mathcal{S}$,

$$\sum_{i\in I}\sum_{j\in J} P(S_i S_j) = \sum_{j\in J}\sum_{i\in I} P(S_j S_i).$$

Then, once more by additivity,

$$\sum_{j\in J} P\left(\sum_{i\in I} S_j S_i\right) = \sum_{j\in J} P(S_j).$$

Therefore, P' is well-defined.

Obviously, $P'(A) \in [0,1]$ and $P'(\Omega) = P(\Omega) = 1$, so it remains to show that P' is σ-additive on $\mathcal{A}(\mathcal{S})$.

σ-additivity of P'. Let $\{A_i\}_{i=1}^{\infty}$ be a sequence of disjoint sets with $A_i \in \mathcal{A}(\mathcal{S})$ for all i, and suppose

$$A := \bigcup_{i=1}^{\infty} A_i \in \mathcal{A}(\mathcal{S}).$$

We need to show that

$$P'\left(\bigcup_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} P'(A_i).$$

Since $A_i \in \mathcal{A}(\mathcal{S})$, write

$$A_i = \sum_{j\in J_i} S_{ij},$$

where J_i is a finite index set and $S_{ij} \in \mathcal{S}$ for all j. Since $A \in \mathcal{A}(\mathcal{S})$, write

$$A = \sum_{k\in K} S_k,$$

where K is a finite index set and $S_k \in \mathcal{S}$. Thus,

$$P'(A) = \sum_{k\in K} P(S_k).$$

As before,

$$S_k = S_k \cap A = S_k \cap \left(\sum_{i=1}^{\infty} A_i\right) = \sum_{i=1}^{\infty} S_k \cap A_i = \sum_{i=1}^{\infty} S_k \cap \left(\sum_{j\in J_i} S_{ij}\right) = \sum_{i=1}^{\infty}\sum_{j\in J_i} S_k \cap S_{ij}.$$

Now $S_k \cap S_{ij} \in \mathcal{S}$, and

$$\sum_{i=1}^{\infty}\sum_{j\in J_i} S_k \cap S_{ij} \in \mathcal{S}.$$

By σ-additivity of P on $\mathcal{S}$,

$$P'(A) = \sum_{k\in K} P(S_k) = \sum_{k\in K} P\left(\sum_{i=1}^{\infty}\sum_{j\in J_i} S_k \cap S_{ij}\right) = \sum_{k\in K}\sum_{i=1}^{\infty}\sum_{j\in J_i} P(S_k S_{ij}) = \sum_{i=1}^{\infty}\sum_{j\in J_i}\sum_{k\in K} P(S_k S_{ij}),$$

and by finite additivity,

$$\sum_{i=1}^{\infty}\sum_{j\in J_i} P(S_{ij}) = \sum_{i=1}^{\infty} P\left(\sum_{j\in J_i} S_{ij}\right) = \sum_{i=1}^{\infty} P(A_i).$$

Combining the two displays,

$$P'(A) = \sum_{i=1}^{\infty} P(A_i) = \sum_{i=1}^{\infty} P'(A_i),$$

which establishes σ-additivity of P' on $\mathcal{A}(\mathcal{S})$. Remains to show uniqueness, so suppose that P_1' and P_2' are two extensions of P to $\mathcal{A}(\mathcal{S})$. Then for any $A \in \mathcal{A}(\mathcal{S})$, we can write $A = \sum_{i\in I} S_i$ where $S_i \in \mathcal{S}$ and I is a finite index set. Then, by the definition of the extension, we have

$$P_1'(A) = \sum_{i\in I} P(S_i) = P_2'(A)$$

which proves that $P_1' = P_2'$ on $\mathcal{A}(\mathcal{S})$ and therefore the extension is unique. □

Theorem 2.4 (Second Extension Theorem)**.** *If P is a σ-additive probability measure on an algebra $\mathcal{A}$, then P has a unique extension to a σ-additive probability measure on the σ-algebra $\sigma(\mathcal{A})$ generated by $\mathcal{A}$.*

Proof. Let $\mathcal{G} = \{\bigcup_{i=1}^{\infty} A_i \,|\, A_i \in \mathcal{A}\}$ be the collection of countable unions of sets in $\mathcal{A}$. If $G \in \mathcal{G}$, then $G = \bigcup_{i=1}^{\infty} A_i = \lim_{n\to\infty} \uparrow B_n, \qquad B_n = \bigcup_{i=1}^{n} A_i \in \mathcal{A}$ for all n. Define $\Pi(G) = \lim_{n\to\infty} \uparrow P(B_n)$. Clearly, $\Pi(A) = P(A), \qquad A \in \mathcal{A}$.

‘‘‘ Some properties of Π are as follows:

(1) $\Omega, \emptyset \in \mathcal{G}$ and

$$\Pi(\Omega) = 1, \qquad \Pi(\emptyset) = 0.$$

(2)

$$0 \le \Pi(G) \le 1 \qquad \text{for all } G \in \mathcal{G}.$$

(3) If $G_1, G_2 \in \mathcal{G}$, then

$$G_1 \cup G_2 \in \mathcal{G}, \qquad G_1 \cap G_2 \in \mathcal{G},$$

and

$$\Pi(G_1 \cup G_2) + \Pi(G_1 \cap G_2) = \Pi(G_1) + \Pi(G_2).$$

Thus, Π is finitely additive on $\mathcal{G}$.

(4) Π is monotone on $\mathcal{G}$.

(5) Π is continuous from below: if $G_n \in \mathcal{G}$ and

$$G_n \uparrow G \in \mathcal{G},$$

then

$$\Pi(G) = \lim_{n\to\infty} \uparrow \Pi(G_n).$$

(6) Π is σ-additive on $\mathcal{G}$.

Thus, after proving these properties, we can extend P on $\mathcal{A}$ to a σ-additive probability measure Π on $\mathcal{G}$.

We now extend Π to $\mathcal{P}(\Omega)$ by defining, for every $A \in \mathcal{P}(\Omega)$,

$$\Pi^*(A) = \inf \{\Pi(G) \,|\, G \in \mathcal{G} \text{ and } A \subseteq G\}.$$

Observe that

$$\Pi^*|_{\mathcal{G}} = \Pi.$$

For every $A_1, A_2 \in \mathcal{P}(\Omega)$, we have

$$\Pi^*(A_1 \cup A_2) + \Pi^*(A_1 \cap A_2) \leq \Pi^*(A_1) + \Pi^*(A_2).$$

Thus, the inequality may be strict or an equality depending on the sets, so we do not have additivity in general, and therefore we cannot conclude σ-additivity.

Define

$$\mathcal{D} = \{D \in \mathcal{P}(\Omega) \,|\, \Pi^*(D) + \Pi^*(D^c) = 1\}.$$

One can prove that $\mathcal{D}$ is a σ-algebra and that Π^* is a probability measure on $(\Omega, \mathcal{D})$. In particular, Π^* is σ-additive on $\mathcal{D}$.

We can see that

$$\mathcal{D} \supseteq \mathcal{A}.$$

By the minimality of $\sigma(\mathcal{A})$, it follows that

$$\mathcal{D} \supseteq \sigma(\mathcal{A}).$$

Moreover,

$$\Pi^*|_{\mathcal{A}} = P.$$

Thus, we have extended P to a σ-additive probability measure on $\sigma(\mathcal{A})$.

Uniqueness follows from Dynkin's π-λ theorem, since $\mathcal{A}$ is a π-system and $\sigma(\mathcal{A})$ is the smallest σ-algebra containing $\mathcal{A}$. "'

□

Definition 2.14 (Lebesgue measure)**.** Begin with intervals of the form $(a, b]$ where $-\infty \leq a \leq b \leq \infty$. Define the semi-algebra $\mathcal{S} = \{(a, b] : -\infty \leq a \leq b \leq \infty\}$ and define $P((a, b]) = b - a$. By the first extension theorem, we can extend P to a probability measure on the algebra generated by $\mathcal{S}$, and by the second extension theorem, we can extend P to a probability measure on the σ-algebra generated by $\mathcal{S}$, which is the Borel σ-algebra. This measure is called the Lebesgue measure. To show additivity, we can write

$$(a, b] = \bigcup_{i=0}^{n-1} (a_i, a_{i+1}]$$

then

$$P((a, b]) = \sum_{i=0}^{n-1} P((a_i, a_{i+1}]) = \sum_{i=0}^{n-1} (a_{i+1} - a_i) = b - a$$

which shows that P is additive on $\mathcal{S}$. To show σ-additivity, we can write

$$(a, b] = \bigcup_{i=0}^{\infty} (a_i, a_{i+1}]$$

then

$$P((a,b]) = \sum_{i=0}^{\infty} P((a_i, a_{i+1}]) = \sum_{i=0}^{\infty}(a_{i+1} - a_i) = b - a$$

which shows that P is σ-additive on $\mathcal{S}$. Alternatively we can use a compactness argument to show that P is σ-additive on $\mathcal{S}$. Begin with a sequence of disjoint intervals $(a_i, b_i]$ such that $\bigcup_{i=0}^{\infty}(a_i, b_i] = (a, b]$. Then, for any n, we have

$$(a,b] \subseteq \bigcup_{i=0}^{n}(a_i, b_i]$$

and therefore

$$b - a = P((a,b]) \leq P\left(\bigcup_{i=0}^{n}(a_i, b_i]\right) = \sum_{i=0}^{n} P((a_i, b_i]) = \sum_{i=0}^{n}(b_i - a_i)$$

which shows that $b - a \leq \sum_{i=0}^{\infty}(b_i - a_i)$. Now, for any $\epsilon > 0$, we can find a finite subcover of the intervals $(a_i - \epsilon/2^i, b_i + \epsilon/2^i)$ that covers $(a, b]$ and therefore we have

$$b-a = P((a,b]) \leq P\left(\bigcup_{i=0}^{n}(a_i - \epsilon/2^i, b_i + \epsilon/2^i)\right) = \sum_{i=0}^{n} P((a_i-\epsilon/2^i, b_i+\epsilon/2^i)) = \sum_{i=0}^{n}(b_i-a_i+\epsilon/2^{i-1})$$

which shows that $b - a \leq \sum_{i=0}^{\infty}(b_i - a_i) + \epsilon$. Since ϵ is arbitrary, we have $b - a \leq \sum_{i=0}^{\infty}(b_i - a_i)$ and therefore P is σ-additive on $\mathcal{S}$. We will denote the Lebesgue measure by λ and we have $\lambda((a,b]) = b - a$.

Remark 2.10. Given a distribution function F, we can construct a probability measure P_F that extends F to all events in $\mathcal{B}$.

$$P_F((-\infty, x]) = F(x)$$

Given $F(x)$ we define the inverse $F^{-1}(y) = \inf\{t \in \mathbb{R} \mid F(t) \geq y\}$.

Remark 2.11. The inverse is left-continuous.

Claim. $F^{-1}(y) > t$ if and only if $y > F(t)$ equivalently $F^{-1}(y) \leq t$ if and only if $y \leq F(t)$.

Definition 2.15. For $A \in \mathbb{R}$ define

$$\zeta_F(A) = \{x \in (0,1] \mid F^{-1}(x) \in A\}$$

Lemma 2.3. *If $A \in \mathcal{B}(\mathbb{R})$, then $\zeta_F(A) \in \mathcal{B}((0,1])$.*

Proof. Define $\mathcal{G} = \{A \subset \mathbb{R} \mid \zeta_F(A) \in \mathcal{B}((0,1])\}$. Now $\mathcal{G}$ contains finite intervals of the form $(a, b]$ because:

$$\begin{aligned}
\zeta_F((a,b]) &= \{x \in (0,1] \mid F^{-1}(x) \in (a,b]\} \\
&= \{x \in (0,1] \mid a < F^{-1}(x) \leq b\} \\
&= \{x \in (0,1] \mid F(a) < x \leq F(b)\} \\
&= (F(a), F(b)]
\end{aligned}$$

One can check that $\mathcal{G}$ is a σ-algebra, and therefore $\mathcal{B}(\mathbb{R}) \subseteq \mathcal{G}$ by minimality of $\mathcal{B}(\mathbb{R})$, which proves the lemma. □

Definition 2.16. Now define $P_F(A) = \lambda(\zeta_F(A))$ for $A \in \mathcal{B}(\mathbb{R})$, where λ is the Lebesgue measure. Then P_F is a probability measure on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$.

Equivalence of the inequalities.

Claim. $A(y) = \{t \in \mathbb{R} \mid F(t) \geq y\}$ is closed

Proof of claim. Let $t_n \downarrow t$ and $t_n \in A(y)$ then $F(t_n) \geq y \quad \forall t_n$ then $F(t) = \lim_{n\to\infty} F(t_n) \geq y$ by right-continuity of F and therefore $t \in A(y)$ Now take $t_n \uparrow t$ and $t_n \in A(y)$ then $F(t_n) \geq y \quad \forall t_n$ then $F(t) = \lim_{n\to\infty} F(t_n) \geq y$ by right-continuity of F and therefore $t \in A(y)$. □

Since $A(y)$ is closed, we have that $F^{-1}(y) = \inf A(y) \in A(y)$ and therefore $F(F^{-1}(y)) \geq y$. Now, if $F^{-1}(y) > t$ then $t < F^{-1}(y) = \inf A(y)$ and therefore $t \notin A(y)$ and therefore $F(t) < y$. Conversely, if $F(t) < y$ then $t \notin A(y)$ and therefore $t < F^{-1}(y) = \inf A(y)$.

□

Chapter 3

Random Variables

3.1 Inverse Maps

Definition 3.1. Let

$$X : \Omega \to \Omega'$$

be a mapping. This induces an inverse map

$$X^{-1} : 2^{\Omega'} \to 2^{\Omega}$$

Proposition 3.1. *Some properties of the inverse map:*

(1) $X^{-1}(\emptyset) = \emptyset$*: let* $\omega \in X^{-1}(\emptyset)$*, then* $X(\omega) \in \emptyset$ *which is a contradiction, therefore* $X^{-1}(\emptyset) = \emptyset$*.*

(2) $X^{-1}(\Omega) = \Omega$*: let* $\omega \in \Omega$*, then* $X(\omega) \in \Omega'$ *and therefore* $\omega \in X^{-1}(\Omega') = X^{-1}(\Omega)$*.*

(3) $X^{-1}(A^c) = (X^{-1}(A))^c$*: let* $\omega \in X^{-1}(A^c)$*, then* $X(\omega) \in A^c$ *and therefore* $\omega \notin X^{-1}(A)$*, which means* $\omega \in (X^{-1}(A))^c$*. Conversely, let* $\omega \in (X^{-1}(A))^c$*, then* $\omega \notin X^{-1}(A)$ *and therefore* $X(\omega) \notin A$*, which means* $X(\omega) \in A^c$ *and thus* $\omega \in X^{-1}(A^c)$*.*

(4) $X^{-1}(\bigcup_{i\in I} A_i) = \bigcup_{i\in I} X^{-1}(A_i)$*: let* $\omega \in X^{-1}(\bigcup_{i\in I} A_i)$*, then* $X(\omega) \in \bigcup_{i\in I} A_i$ *and therefore* $X(\omega) \in A_i$ *for some* $i \in I$*, which means* $\omega \in X^{-1}(A_i)$ *for some* $i \in I$ *and thus* $\omega \in \bigcup_{i\in I} X^{-1}(A_i)$*. Conversely, let* $\omega \in \bigcup_{i\in I} X^{-1}(A_i)$*, then* $\omega \in X^{-1}(A_i)$ *for some* $i \in I$ *and therefore* $X(\omega) \in A_i$ *for some* $i \in I$*, which means* $X(\omega) \in \bigcup_{i\in I} A_i$ *and thus* $\omega \in X^{-1}(\bigcup_{i\in I} A_i)$*.*

(5) $X^{-1}(\bigcap_{i\in I} A_i) = \bigcap_{i\in I} X^{-1}(A_i)$*: By De Morgan's of (4).*

Proposition 3.2. *If* $\mathcal{B}$ *is a* σ*-algebra on* Ω'*, then* $X^{-1}(\mathcal{B}) = \{X^{-1}(B) : B \in \mathcal{B}\}$ *is a* σ*-algebra on* Ω*.*

Proof. Let $\mathcal{B}$ be a σ-algebra on Ω' and X^{-1} be an inverse map from $(\Omega', \mathcal{B})$ to $(\Omega, \mathcal{F})$.

- We have $\Omega \in X^{-1}(\mathcal{B})$ since $\Omega' \in \mathcal{B}'$ and $X^{-1}(\Omega') = \Omega$ by axioms
- If $A \in \mathcal{B}$ then $A^c \in \mathcal{B}$ and therefore $X^{-1}(A^c) = (X^{-1}(A))^c \in X^{-1}(\mathcal{B})$ by axioms
- If $A_i \in \mathcal{B}$ for $i \in I$ where I is countable index set then $\bigcup_{i\in I} A_i \in \mathcal{B}$ and therefore $X^{-1}(\bigcup_{i\in I} A_i) = \bigcup_{i\in I} X^{-1}(A_i) \in X^{-1}(\mathcal{B})$ by axioms

□

Proposition 3.3. *If* C' *is a class of sets in* Ω'*, then* $X^{-1}(\sigma(C')) = \sigma(X^{-1}(C'))$*.*

Proof. Obviously $\sigma(X^{-1}(C')) \subseteq X^{-1}(\sigma(C'))$ by minimality. The other direction is as follows, define

$$\mathcal{F} = \{B' \in \mathcal{P}(\Omega') : X^{-1}(B') \in \sigma(X^{-1}(C'))$$

Then $\mathcal{F}$ is a σ-algebra (one can check this). Now $X^{-1}(\mathcal{F}) \subseteq \sigma(X^{-1}(C'))$ by definition of $\mathcal{F}$ and $C' \subseteq \mathcal{F}$ by definition of $\mathcal{F}$, therefore $\sigma(C') \subseteq \mathcal{F}$ by minimality and therefore $X^{-1}(\sigma(C')) \subseteq X^{-1}(\mathcal{F}) \subseteq \sigma(X^{-1}(C'))$. □

3.2 Measurable Maps

Definition 3.2. Let $X : (\Omega, \mathcal{F}) \to (\Omega', \mathcal{F}')$ be a mapping. We say that X is measurable if

$$X^{-1}(\mathcal{F}') \subseteq \mathcal{F}$$

In this case, we say that X is a random element and by abuse of notation we write $X \in \mathcal{F} \setminus \mathcal{F}'$. If $\Omega' = \mathbb{R}$ and $\mathcal{F}' = \mathcal{B}(\mathbb{R})$, then we say that X is a random variable. If $\Omega' = \mathbb{R}^k$ and $\mathcal{F}' = \mathcal{B}(\mathbb{R}^k)$, then we say that X is a random vector. If $\Omega' = \mathbb{R}^\infty$ and $\mathcal{F}' = \mathcal{B}(\mathbb{R}^\infty)$, then we say that X is a random sequence.

Definition 3.3. Let $(\Omega, \mathcal{B}, P)$ be a probability space and suppose that $X : (\Omega, \mathcal{B}) \to (\Omega', \mathcal{B}')$ be a measurable map. Define for $A' \subseteq \Omega'$

$$[X \in A'] = X^1(A') = [\omega : X(\omega) \in A']$$

Furthermore, define the set function $P \circ X^{-1}$ on $(\Omega', \mathcal{B}')$ to be the *induced probability* or the *distribution* of X. To verify it is indeed a measure on $\mathcal{B}'$ one can show that:

(1) $P \circ X^{-1}(\Omega') = P(\Omega) = 1$

(2) $P \circ X^{-1}(A') \geq 0 \quad \forall A' \in \mathcal{B}'$

(3) $\sigma-$additivity.

Usually we will write $P \circ X^{-1}(A') = P[X \in A']$

Remark 3.1. If X is a r.v then $P \circ X^{-1}$ is the measure induced on $\mathbb{R}$ by the distribution function

$$P \circ X^{-1}((-\infty, x]) = P[X \leq x]$$

Proposition 3.4 (Test for Measurability)**.** *Suppose $X : \Omega \to \Omega'$ where $(\Omega, \mathcal{B}), (\Omega', \mathcal{B}')$ are measurable spaces. Suppose that C' generates $\mathcal{B}'$ that is $\mathcal{B}' = \sigma(C')$ then X is measurable if and only if $X^{-1}(C') \subseteq \mathcal{B}'$*

Proof. $X^{-1}(C') \subseteq \mathcal{B}$ then by minimality we get that $\sigma(X^{-1}(C') \subseteq \mathcal{B})$ but $\sigma(X^{-1}(C')) = X^{-1}(\sigma(C'))$ then $X^{-1}(\sigma(C')) \subseteq \mathcal{B}$ □

Corollary 3.1. *If $X : \Omega \to \mathbb{R}$ is an r.v if and only if $X^{-1}((-\infty.x]) = [X \leq x] \in \mathcal{B}\ \forall x \in \mathbb{R}$*

Proof. This follows directly from the fact that $\sigma($ open sets of $\mathbb{R}) = \mathcal{B}(\mathbb{R})$ □

3.3 Composition

Proposition 3.5. *Let X_1, X_2 be two measurable maps defined as follows:*

$$X_1 : (\Omega_1, \mathcal{B}_1) \to (\Omega_2, \mathcal{B}_2)$$

and

$$X_2 : (\Omega_2, \mathcal{B}_2) \to (\Omega_3, \mathcal{B}_3)$$

where the $(\Omega_i, \mathcal{B}_i)$ are measurable spaces. Define $X_2 \circ X_1 : (\Omega_1, \mathcal{B}_1) \to (\Omega_3, \mathcal{B}_3)$ by

$$(X_2 \circ X_1)(\omega) = X_2(X_1(\omega))$$

Then $X_2 \circ X_1 \in \mathcal{B}_1/\mathcal{B}_3$.

Proof. — Exercise □

Proposition 3.6. *Suppose we have 2 metric spaces (S_i, d_i) for $i = 1, 2$. Let the borel $\sigma-$field be $\mathcal{S}_i$ if $X : S_1 \to S_2$ is continuous then X is measurable: $X \in \mathcal{S}_1/\mathcal{S}_2$.*

Proof. Let O_i be the class of open substes of S_i. If X is continuous then the inverse images of open sets are open i,e

$$X^{-1}(O_2) \subseteq O_1 \subseteq \sigma(O_1) = S_1$$

so $X \in \mathcal{S}_1/\mathcal{S}_2$ by the test for measurability. □

Corollary 3.2. *If $X = (X_1, \ldots, X_k)$ is a random vector and*

$$g : \mathbb{R}^k \to \mathbb{R}$$

then $g(X) = g(X_1, \ldots, X_k)$ is a random variable, in particular if g is continuous then g is measurable and therefore $g(X)$ is a random variable.

Example 3.1. Some examples are:

(1) Component sum: $g(x_1, \ldots, x_k) = \sum_{i=1}^k x_i$

(2) Component product: $g(x_1, \ldots, x_k) = \prod_{i=1}^k x_i$

(3) Component max: $g(x_1, \ldots, x_k) = \max\{x_1, \ldots, x_k\}$

(4) Component min: $g(x_1, \ldots, x_k) = \min\{x_1, \ldots, x_k\}$

(5) Component average: $g(x_1, \ldots, x_k) = \frac{1}{k} \sum_{i=1}^k x_i$

(6) Component sum of squares: $g(x_1, \ldots, x_k) = \sum_{i=1}^k x_i^2$

(7) Projection map $g = \pi_i : \mathbb{R}^k \to \mathbb{R}$ defined by $\pi_i(x_1, \ldots, x_k) = x_i$ then $\pi_i(X) = X_i$ is a random variable.

Proposition 3.7. *$X = (X_1, \ldots, X_k)$ is a random vector, i.e a measurable map form $(\Omega, \mathcal{F})$ to $(\mathbb{R}^k, \mathcal{B}(\mathbb{R}^k))$ if and only if X_i is a random variable for all $i = 1, \ldots, k$.*

Proposition 3.8. *$X = (X_1, X_2, \ldots)$ is a random sequence, i.e a measurable map from $(\Omega, \mathcal{F})$ to $(\mathbb{R}^\infty, \mathcal{B}(\mathbb{R}^\infty))$ if and only if X_i is a random variable for all $i = 1, 2, \ldots$ and $\{X_1, \ldots, X_n\}$ is a random vector for any n.*

3.4 Measurability and limits

Proposition 3.9. *Let $X_1, X_2, \ldots$ be random variables defined on $(\Omega, \mathcal{B})$. Then*

(1) $\inf_n X_n$ *and* $\sup_n X_n$ *are a random variable*

(2) $\limsup_{n\to\infty} X_n$ *and* $\liminf_{n\to\infty} X_n$ *are a random variable*

(3) $\lim_{n\to\infty} X_n$ *is a random variable if the limit exists.*

(4) The set on which $\{X_n\}$ has a limit is measurable; that is $\{\omega : \lim_{n\to\infty} X_n(\omega)$ exists$\} \in \mathcal{B}$.

Proof. We will prove the previous items:

(1) We have that $[\inf_n X_n \leq x] = \cup_n [X_n \leq x] \in \mathcal{B}$ and $[\sup_n X_n \leq x] = \cap_n [X_n \leq x] \in \mathcal{B}$.

(2) We have by (1) that $\inf_n X_n$ is a r.v and that $\sup_n X_n$ is an r.v then $\limsup_n = \inf(\sup X_n)$ is a r.v and $\liminf_n = \sup(\inf X_n)$ is a r.v.

(3) If the limit exists then $\lim_n X_n = \limsup_n X_n = \liminf_n X_n$ and therefore $\lim_n X_n$ is a r.v.

(4) Let $\mathbb{Q}$ be the set of all rational real numbers, then from any analysis course we know that $\mathbb{Q}$ is countable. We have

$$\begin{aligned}
\{\omega : \lim_{n\to\infty} X_n(\omega) \text{ exists}\}^c &= \{\omega : \liminf_{n\to\infty} X_n(\omega) < \limsup_{n\to\infty} X_n(\omega)\} \\
&= \bigcup_{r\in\mathbb{Q}} \{\liminf_{n\to\infty} X_n(\omega) \leq r < \limsup_{n\to\infty} X_n(\omega)\} \\
&= \bigcup_{r\in\mathbb{Q}} \{\liminf_{n\to\infty} X_n(\omega) \leq r\} \cap \{\limsup_{n\to\infty} X_n(\omega) > r\} \\
&= \bigcup_{r\in\mathbb{Q}} \{\liminf_{n\to\infty} X_n(\omega) \leq r\} \cap \{\limsup_{n\to\infty} X_n(\omega) \leq r\}^c \in \mathcal{B}
\end{aligned}$$

Since $[\liminf X_n \leq r] \in \mathcal{B}$ and $[\limsup X_n \leq r] \in \mathcal{B}$ by (1) and $\mathcal{B}$ is closed under complements and countable unions, we have that the set on which $\{X_n\}$ has a limit is measurable.

□

3.5 σ − field Generated by Maps

Definition 3.4. Let X be a r.v. The σ − field generated by X denoted as $\sigma(X)$ is defined as

$$\sigma(X) = X^{-1}(\mathcal{B}(\mathbb{R}))$$

Another equivalent description is

$$\sigma(X) = \{[X \in B] : B \in \mathcal{B}(\mathbb{R})\}$$

which is the collection of all events of the form $[X \in B]$ for $B \in \mathcal{B}(\mathbb{R})$. More generally, suppose that

$$X : (\Omega, \mathcal{B}) \to (\Omega', \mathcal{B}')$$

then we define

$$\sigma(X) = X^{-1}(\mathcal{B}')$$

. If $\mathcal{F}$ is a sub σ − field of $\mathcal{B}$, we say that X is measurable with respect to $\mathcal{F}$ written $X \in \mathcal{F}$ if $\sigma(X) \subseteq \mathcal{F}$. If for each t in some index set T we have $X_t : (\Omega, \mathcal{B}) \to (\Omega', \mathcal{B}')$ then we denote by

$$\sigma(X_t, t \in T) = \vee_{t\in T} \sigma(X_t)$$

Chapter 4

Independence

Definition 4.1 (Independence of 2 events)**.** Suppose we have a probability space $(\Omega, \mathcal{B}, P)$ we say that two events $A, B \in \mathcal{B}$ are independent if

$$P(A \cap B) = P(A)P(B)$$

Definition 4.2 (Independence of finitely many events)**.** The events $A_i \quad i = 1, \ldots, n$ are independent if

$$P\left(\bigcap_i A_i\right) = \prod_i P(A_i)$$

Definition 4.3 (Independence classes)**.** Let $C_i \subseteq \mathcal{B}$ with $i = 1, \ldots, n$. The classes C_i are independent if for any choice $A_1, \ldots, A_n$ with each $A_i \in C_i$ we have the events independent.

Theorem 4.1 (Basic Criterion)**.** *If for each $i = 1, \ldots, n$, C_i is a non-empty class of events satisfying the following:*

(1) C_i is a $\pi-$system

(2) The classes C_i are independent

Then

$$\sigma(C_1), \ldots, \sigma(C_n)$$

are independent.

Proof. We will prove this for $n = 2$ then one may apply induction. Let C_1, C_2 be two $\pi-$systems that are independent. Define

$$\mathcal{D}_1 = \{A \in \sigma(C_1) : P(A \cap B) = P(A)P(B) \quad \forall B \in C_2\}$$

We will show that $\mathcal{D}_1$ is a $\lambda-$system. First we have $\Omega \in \mathcal{D}_1$ since $P(\Omega \cap B) = P(B) = P(\Omega)P(B)$ for all $B \in C_2$. Now let $A_1, A_2 \in \mathcal{D}_1$ with $A_1 \subseteq A_2$ then for all $B \in C_2$ we have:

(1) $P(A_2 \setminus A_1 \cap B) = P(A_2 \cap B) - P(A_1 \cap B) = P(A_2)P(B) - P(A_1)P(B) = P(A_2 \setminus A_1)P(B)$

(2) Let $A_n \in \mathcal{D}_1$ be disjoint then for all $B \in C_2$ we have:

$$P\left(\bigcup_n A_n \cap B\right) = \sum_n P(A_n \cap B) = \sum_n P(A_n)P(B) = P\left(\bigcup_n A_n\right) P(B)$$

Therefore $\mathcal{D}_1$ is a $\lambda-$system. Now since $C_1 \subseteq \mathcal{D}_1$ and C_1 is a $\pi-$system, by Dynkin's $\pi - \lambda$ theorem we have that $\sigma(C_1) \subseteq \mathcal{D}_1$ and therefore for all $A \in \sigma(C_1)$ and $B \in C_2$ we have $P(A \cap B) = P(A)P(B)$. Now we can define

$$\mathcal{D}_2 = \{B \in \sigma(C_2) : P(A \cap B) = P(A)P(B) \quad \forall A \in \sigma(C_1)\}$$

and by the same argument we have that $\mathcal{D}_2$ is a $\lambda-$system and $C_2 \subseteq \mathcal{D}_2$ and therefore $\sigma(C_2) \subseteq \mathcal{D}_2$ and therefore for all $A \in \sigma(C_1)$ and $B \in \sigma(C_2)$ we have $P(A \cap B) = P(A)P(B)$ which proves the theorem for $n = 2$. The general case follows by induction □

Definition 4.4. Let T be an arbitrary index set. The classes are independent families if for each finite $I \subseteq T$ $C_t, t \in I$ are independent.

Corollary 4.1. *If $\{C_t, t \in T\}$ are non-empty $\pi-$systems that are indepented then $\{\sigma(C_t) : t \in T\}$ are independent.*

Proof. From the basic criterion. □

4.1 Independent Random Variables

Definition 4.5. $\{X_t, t \in T\}$ is an independent family of random variables if $\{\sigma(X_t), t \in T\}$ are independent $\sigma -$ field.

Definition 4.6. For a family of random variables $\{X_t, t \in T\}$ we define the joint distribution function

$$F_J(x_t, t \in J) = P[X_t \leq x_t, t \in J]$$

for any finite $J \subseteq T$.

Theorem 4.2. *A family of random variables $\{X_t, t \in T\}$ is independent if for all finite $J \subseteq T$ we have*

$$F_J(x_t, t \in J) = \prod_{t \in J} P[X_t \leq x_t] \quad \forall x_t \in \mathbb{R}$$

Proof. It suffices to show for a finite index set J that $\{x_t, t \in T\}$ is independent if and only if the above holds. Define

$$C_t = \{[X_t \leq x], x \in \mathbb{R}\}$$

then

(1) C_t is a $\pi-$system since $[X_t \leq x_1] \cap [X_t \leq x_2] = [X_t \leq \min(x_1, x_2)]$

(2) $\sigma(C_t) = \sigma(X_t)$

So by basic criterion we are done. □

Corollary 4.2. *The finite collection of random variables $X_1, \ldots, X_n$ is independent if and only if*

$$P[X_1 \leq x_1, \ldots, X_n \leq x_n] = \prod_{i=1}^{n} P[X_i \leq x_i]$$

for all $x_1, \ldots, x_n \in \mathbb{R}$.

Definition 4.7. We say a random variable is discrete if it has a countable range.

Corollary 4.3. *Let $X_1, \ldots, X_n$ be discrete random variables with range R. Then $X_1, \ldots, X_n$ are independent if and only if*

$$P[X_1 = x_1, \ldots, X_n = x_n] = \prod_{i=1}^{n} P[X_i = x_i]$$

for all $x_1, \ldots, x_n \in R$

4.2 Borel-Cantelli Lemma

Proposition 4.1 (Borel-Cantelli Lemma)**.** *Let $\{A_n\}$ be any events. If $\sum_n P(A_n) < \infty$ then $P(A_n \text{ i.o}) = P(\limsup_n A_n) = 0$.*

Proof. We have

$$\begin{aligned} P(A_n \text{ i.o}) &= P(\lim_{n\to\infty} \cup_{j\geq n} A_j) \\ &= \lim_{n\to\infty} P(\cup_{j\geq n} A_j) \\ &\leq \limsup_{n\to\infty} \sum_{j\geq n} P(A_j) \\ &= 0 \end{aligned}$$

□

Proposition 4.2 (Borel Zero-One Law)**.** *Let $\{A_n\}$ be independent events. Then* $P(A_n \text{ i.o}) = \begin{cases} 0 & \text{if } \sum_n P(A_n) < \infty \\ 1 & \text{if } \sum_n P(A_n) = \infty \end{cases}$

Proof. Borel Cantelli takes care of the first case. Conversely suppose we have divergence. Then

$$\begin{aligned} P(A_n \text{ i.o}) &= P(\lim_{n\to\infty} \cup_{j\geq n} A_j) \\ &= \lim_{n\to\infty} P(\cup_{j\geq n} A_j) \\ &= 1 - \lim_{n\to\infty} P(\cap_{j\geq n} A_j^c) \\ &= 1 - \lim_{n\to\infty} \prod_{j\geq n} P(A_j^c) \\ &= 1 - \lim_{n\to\infty} \prod_{j\geq n} (1 - P(A_j)) \\ &\geq 1 - \lim_{n\to\infty} e^{-\sum_{j\geq n} P(A_j)} = 1 \end{aligned}$$

□

4.3 Kolmogorov Zero-One Law

Definition 4.8 (Tail $\sigma-$field)**.** Let $\{X_n\}$ be a sequence of random variables. The tail $\sigma-$field is defined as

$$\mathcal{T} = \bigcap_{n=1}^{\infty} \sigma(X_{n+1}, \ldots) = \lim \downarrow \sigma(X_n, X_{n+1}, \ldots)$$

The events in $\mathcal{T}$ are called tail events.

Example 4.1. Some examples of Tail $\sigma-$fields are:

(1) $\{\omega : \sum_n X_n(\omega) < \infty\} \in \mathcal{T}$

(2) $\{\omega : \lim_n X_n(\omega) \text{ exists}\} \in \mathcal{T}$

Theorem 4.3 (Kolmogorov Zero-One Law)**.** *If $\{X_n\}$ are independent random variables with a tail $\sigma-$field T then $\Lambda \in T$ implies that $P(\Lambda) = 0$ or 1 so that the tail $\sigma-$field is almost trivial.*

Lemma 4.1. *Let $\mathcal{G}$ be an almost trivial $\sigma-$field and let X be a random variable measurable with respect to $\mathcal{G}$. Then there exists c such that $P(X = c) = 1$.*

Proof of Lemma. Let $F(x) = P(X \leq x)$ then F is nondecreasing and since $[X \leq x] \in \sigma(X) \subseteq \mathcal{G}$ we have $F(x) = 0$ or 1 for all x. Let $c = \sup\{x : F(x) = 0\}$ then $F(c) = 1$ and $F(c-) = 0$ and therefore $P(X = c) = 1$. □

Corollary 4.4 (Corollaries of Kolmogorov Zero-One Law)**.** *Let $\{X_n\}$ be independent random variables then the following are true:*

(1) The event $\{\sum_n X_n \text{ converges}\} \in \mathcal{T}$ and therefore $P(\sum_n X_n \text{ converges}) = 0$ or 1.

(2) The random variables $\limsup X_n$ and $\liminf X_n$ are constant with probability 1.

(3) The event $\{\omega : S_n(\omega)/n \to 0\}$ has probability 0 or 1.

Proof of Kolmogorov Zero-One Law. Suppose that $\Lambda \in T$. We show that Λ is independent to itself i.e $P(\Lambda \cap \Lambda) = P(\Lambda)P(\Lambda)$ which implies that $P(\Lambda) = 0$ or 1. To show independence, define

$$F_n = \sigma(X_1, \ldots, X_n) = \vee_{j=1}^n \sigma(X_j)$$

So that $F_n \uparrow$ and

$$F_\infty = \vee_{n=1}^\infty F_n$$

Note that $\Lambda \in T \subseteq F_n' = \sigma(X_{n+1}, X_{n+2}, \ldots) \subseteq F_\infty$. Now for all n we have $\Lambda \in F_n'$ so since $F_n \perp F_n'$ we have that Λ is independent of F_n and therefore Λ is independent of F_∞. Since $\Lambda \in F_\infty$ we have that Λ is independent of itself and therefore $P(\Lambda) = 0$ or 1. □

Chapter 5

Integration and Expectation

5.1 Simple functions

Definition 5.1. Recall that a function on a probability space $(\Omega, \mathcal{B}, P)$

$$X : \Omega \to \mathbb{R}$$

is said to be simple if it has a finite range. This means that a simple function is $\mathcal{B}/\mathcal{B}(R)$ measurable. Such a function can be written in the form

$$X(\omega) = \sum_{i=1}^{k} a_i \mathbb{I}_{A_i}$$

where $a_i \in \mathbb{R}$ and $A_i \in \mathcal{B}$ and A_i are disjoint and $\bigcup_{i=1}^{k} A_i = \Omega$. Recall that $\sigma(X) = \sigma(A_i : i = 1, \ldots, k) = \left\{\bigcup_{i \in I} A_i : I \subseteq \{1, \ldots, k\}\right\}$. Let $\mathcal{E}$ be the set of all simple functions on Ω. We have the following properties of $\mathcal{E}$:

(1) $\mathcal{E}$ is a vector space.

(2) If $X, Y \in \mathcal{E}$ then $XY \in \mathcal{E}$.

(3) If $X \in \mathcal{E}$ then $X \wedge Y, X \vee Y \in \mathcal{E}$ for any $Y \in \mathcal{E}$.

5.2 Measurability and Simple functions

Theorem 5.1. *Suppose that $X(\omega) \geq 0$ for all ω then $X \in \mathcal{B}/\mathcal{B}(\mathbb{R})$ if and only if there exists simple functions $X_n \in \mathcal{E}$ such that $X_n(\omega) \uparrow X(\omega)$ for all ω.*

Proof. If $X_n \in \mathcal{E}$ then $X_n \in \mathcal{B}/\mathcal{B}(\mathbb{R})$ and if $X = \lim \uparrow X_n$ then $X \in B/\mathcal{B}(\mathbb{R})$ by measurability and limits. Conversely, suppose that $0 \leq X \in \mathcal{B}/\mathcal{B}(\mathbb{R})$. Define

$$X_n = \sum_{k=1}^{n2^n} \mathbb{I}_{[\frac{k-1}{2^n}, \frac{k}{2^n})} + n\mathbb{I}_{[X \geq n]}$$

Since $X \in \mathcal{B}/\mathcal{B}(\mathbb{R})$ we have that $X_n \in \mathcal{E}$ adn $X_n \leq X_{n+1}$ and if $X(\omega) < \infty$ then for all large enough n we have

$$|X(\omega) - X_n(\omega)| \leq \frac{1}{2^n} \to 0$$

If $X(\omega) = \infty$ then $X_n(\omega) = n \to \infty$ and therefore $X_n(\omega) \uparrow X(\omega)$ for all ω. □

5.3 Expectation and Integration

Definition 5.2. Suppose that X is a simple random variable of the form

$$X = \sum_{i=1}^{n} a_i \mathbb{I}_{A_i}$$

where $|a_i| < \infty$ and $\cup A_i = \Omega$. We define the expectation of X to be

$$\mathbb{E}(X) = \int X dP = \sum_{i=1}^{k} a_i P(A_i)$$

Some properties are as follows:

(1) $\mathbb{E}(1) = 1$ and $\mathbb{E}(c) = c$ for any constant c.

(2) $\mathbb{E}(\mathbb{I}_(A_i)) = P(A_i)$.

(3) If $X \geq 0$ and simple then $\mathbb{E}(X) \geq 0$.

(4) The expectation is linear, that is $\mathbb{E}(aX + bY) = a\mathbb{E}(X) + b\mathbb{E}(Y)$ for any simple random variables X, Y and constants a, b.

(5) The expectation on simple random variables is monotone, that is if $X \leq Y$ then $\mathbb{E}(X) \leq \mathbb{E}(Y)$.

(6) If $X_n, X \in \mathcal{E}$ then either $X_n \uparrow X$ or $X_n \downarrow X$ then $\mathbb{E}(X_n) \uparrow \mathbb{E}(X)$ or $\mathbb{E}(X_n) \downarrow \mathbb{E}(X)$ respectively.

Definition 5.3. We will define $\mathcal{E}_+$ and $\overline{\mathcal{E}_+}$ as the sets of all non-negative simple functions and all bounded simple functions, respectively. These sets are as follows:

(1) $\mathcal{E}_+ = \{X \in \mathcal{E} : X(\omega) \geq 0\} = \{X : (\Omega, \mathcal{B}) \to (\mathbb{R}_+, \mathcal{B}(\mathbb{R}_+))\}$

(2) $\overline{\mathcal{E}_+} = \{X \in \mathcal{E} : |X(\omega)| < M < \infty\} = \{X : (\Omega, \mathcal{B}) \to (\overline{\mathbb{R}_+}, \mathcal{B}(\overline{\mathbb{R}_+}))\}$

Proposition 5.1 (Well Definition)**.** *The expectation is well defined on $\overline{\mathcal{E}_+}$, since if $X_n \in \mathcal{E}_+$ and $Y_m \in \mathcal{E}_+$ and $X_n \uparrow X$ and $Y_m \uparrow X$ then $\lim_n \mathbb{E}(X_n) = \lim_m \mathbb{E}(Y_m)$.*

Proof. Because of symmetric roles its enough to show that if $X_n, Y_m \in \mathcal{E}_+$ and both sequences $\{X_n\}, \{Y_m\}$ are non decreasing then the assumption

$$\lim_{n\to\infty} \uparrow X_n \leq \lim_{m\to\infty} \uparrow Y_m \implies \lim \uparrow \mathbb{E}(X_n) \leq \lim \uparrow \mathbb{E}(Y_m)$$

To prove this we note that as $m \to \infty$ we have $X_n \wedge Y_m \uparrow X_n$ and therefore $\lim_{m\to\infty} \mathbb{E}(X_n \wedge Y_m) = \mathbb{E}(X_n)$. Now since $X_n \wedge Y_m \leq Y_m$ we have that $\mathbb{E}(X_n \wedge Y_m) \leq \mathbb{E}(Y_m)$ and therefore $\mathbb{E}(X_n) = \lim_{m\to\infty} \mathbb{E}(X_n \wedge Y_m) \leq \lim_{m\to\infty} \mathbb{E}(Y_m)$ and taking the limit as $n \to \infty$ we get the desired result. □

Theorem 5.2 (Basic Properties)**.** *The expectation on $\overline{\mathcal{E}_+}$ has the following basic properties:*

(1) $0 \leq \mathbb{E}(X) \leq \infty$

(2) $\mathbb{E}$ *is linear, that is* $\mathbb{E}(aX + bY) = a\mathbb{E}(X) + b\mathbb{E}(Y)$ *for any* $X, Y \in \overline{\mathcal{E}_+}$ *and constants* a, b.

(3) Monotone Convergence Theorem: If $0 \leq X_n \uparrow X$ *then* $\mathbb{E}(X_n) \uparrow \mathbb{E}(X)$.

Proof of Monotone Convergence Theorem. Let $X_n \in \overline{\mathcal{E}_+}$ and $X_n \uparrow X$ then by the well definition of expectation we have that $\lim_{n\to\infty} \mathbb{E}(X_n) = \mathbb{E}(X)$. □

Corollary 5.1 (Series version of Monotone Convergence Theorem)**.** *If $X_n \in \overline{\mathcal{E}_+}$ then*

$$\mathbb{E}\left(\sum_{n=1}^{\infty} X_n\right) = \sum_{n=1}^{\infty} \mathbb{E}(X_n)$$

Proof. Let $S_n = \sum_{i=1}^{n} X_i$ then $S_n \uparrow \sum_{n=1}^{\infty} X_n$ and therefore by the monotone convergence theorem we have that

$$\mathbb{E}\left(\sum_{n=1}^{\infty} X_n\right) = \lim_{n\to\infty} \mathbb{E}(S_n) = \lim_{n\to\infty} \sum_{i=1}^{n} \mathbb{E}(X_i) = \sum_{n=1}^{\infty} \mathbb{E}(X_n)$$

□

Theorem 5.3 (Fatou's Lemma)**.** *If $X_n \geq 0$ then*

$$\mathbb{E}(\liminf_{n\to\infty} X_n) \leq \liminf_{n\to\infty} \mathbb{E}(X_n)$$

More generally there exists $Z \in \ell_1$ and $X_n \geq Z$ then

$$\mathbb{E}(\liminf_{n\to\infty} X_n) \leq \liminf_{n\to\infty} \mathbb{E}(X_n)$$

Proof. Let $Y_n = \inf_{k\geq n} X_k$ then $Y_n \uparrow \liminf_{n\to\infty} X_n$ and therefore by the monotone convergence theorem we have that

$$\mathbb{E}(\liminf_{n\to\infty} X_n) = \lim_{n\to\infty} \mathbb{E}(Y_n)$$

Now since $Y_n \leq X_n$ we have that $\mathbb{E}(Y_n) \leq \mathbb{E}(X_n)$ and therefore $\lim_{n\to\infty} \mathbb{E}(Y_n) \leq \liminf_{n\to\infty} \mathbb{E}(X_n)$ which proves the statement. □

Corollary 5.2 (More Fatou)**.** *If $X_n \leq Z$ where $Z \in \ell_1$ then*

$$\mathbb{E}(\limsup_{n\to\infty} X_n) \geq \limsup_{n\to\infty} \mathbb{E}(X_n)$$

Theorem 5.4 (Dominated Convergence Theorem)**.** *If $X_n \to X$ almost surely and $|X_n| \leq Z$ where $\mathbb{E}(Z) < \infty$, then*

$$\mathbb{E}(X_n) \to \mathbb{E}(X).$$

Proof. By Fatou's lemma,

$$\begin{aligned}\mathbb{E}(X) &= \mathbb{E}\left(\liminf_{n\to\infty} X_n\right) \\ &\leq \liminf_{n\to\infty} \mathbb{E}(X_n).\end{aligned}$$

Applying Fatou's lemma to $Z - X_n$, we get

$$\begin{aligned}\mathbb{E}(Z - X) &\leq \liminf_{n\to\infty} \mathbb{E}(Z - X_n) \\ &= \mathbb{E}(Z) - \limsup_{n\to\infty} \mathbb{E}(X_n).\end{aligned}$$

Hence,

$$\limsup_{n\to\infty} \mathbb{E}(X_n) \leq \mathbb{E}(X).$$

Therefore,

$$\begin{aligned}\mathbb{E}(X) &\leq \liminf_{n\to\infty} \mathbb{E}(X_n) \\ &\leq \limsup_{n\to\infty} \mathbb{E}(X_n) \\ &\leq \mathbb{E}(X),\end{aligned}$$

which implies that $\mathbb{E}(X_n) \to \mathbb{E}(X)$. □

5.4 Indefinite Integrals

Definition 5.4. If $X \in \ell_1$ then we define

$$\int_A X dP = \mathbb{E}(X 1_A)$$

We call X the integrand.

Proposition 5.2. *Suppose that $X \geq 0$. For postiive integrands then we have the following properties:*

(1) $0 \leq \int_A X dP \leq \mathbb{E}(X)$

(2) $\int_A X dP = 0$ *if and only if* $P(A \cap [X > 0]) = 0$

(3) If $\{A_n : n \geq 1\}$ *is a sequence of disjoint events then*

$$\int_{\cup_n A_n} X dP = \sum_n \int_{A_n} X dP$$

5.5 Transformation Theorem and Densities

Definition 5.5. Given 2 measurable spaces $(\Omega, \mathcal{B})$ and $(\Omega', \mathcal{B}')$ and a measurable map $T : (\Omega, \mathcal{B}) \to (\Omega', \mathcal{B}')$ and a measure P on $\mathcal{B}$. Define $P' = P \circ T^{-1}$ on $\mathcal{B}'$ by $P'(A') = P(T^{-1}(A'))$ for all $A' \in \mathcal{B}'$.

Theorem 5.5. *Suppose $X' : (\Omega', \mathcal{B}') \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ is a random variable with domain Ω' then $X = X' \circ T$ is a random variable with domain Ω.*

(1) If $X' \geq 0$ then

$$int_\Omega X'(T(\omega))P(d\omega) = \int_{\Omega'} X'(\omega')P'(d\omega')$$

where $P' = P \circ T^{-1}$.

(2) If $X' \in \ell_1(P')$ if and only if $X' \circ T \in \ell_1(P)$. In which case

$$\int_{T^{-1}(A')} X'(T(\omega))d(P\omega) = \int_{A'} X'(\omega')P'(d\omega')$$

5.6 Densities

Definition 5.6. Let X be a random vector on $(\Omega, \mathcal{B}, P)$ with distribution F. We say X or F is absolutely continuous if there exists a non-negative measurable function $f : \mathbb{R}^k \to \mathbb{R}_+$ such that for all $A \in \mathcal{B}$,

$$F(A) = \int_A f(x)dx.$$

Where dx is the Lebesgue measure on $\mathbb{R}^k$. The function f is called the density of X or F.

Proposition 5.3. *Let $g : \mathbb{R}^k \to \mathbb{R}_+$ be a non-negative measurable function. Suppose that X is random vector with distribution F. If F is absolutely continuous with densitiy f then we have for the expectation of $g(X)$,*

$$\mathbb{E}(g(X)) = \int_{\mathbb{R}^k} g(x)f(x)dx$$

5.7 Product Spaces

Definition 5.7. Let Ω_1, Ω_2 be two sets. Define the product space:

$$\Omega_1 \times \Omega_2 = \{(\omega_1, \omega_2) : \omega_i \in \Omega_i : i = 1, 2\}$$

and define the projection map by

$$\pi_i(\omega_1, \omega_2) = \omega_i$$

such that

$$\pi_i : \Omega_1 \times \Omega_2 \to \Omega_i$$

If $A \subseteq \Omega_1 \times \Omega_2$ define

$$A_{\omega_1} = \{\omega_2 : (\omega_1, \omega_2) \in A\} \subseteq \omega_2$$

and

$$A_{\omega_2} = \{\omega_1 : (\omega_1, \omega_2) \in A\} \subseteq \omega_1$$

We call A_{ω_i} the section of A at ω_i.

Proposition 5.4. *Some properties of set sections are*

(1) If $A \subseteq \Omega_1 \times \Omega_2$ then $(A^c)_{\omega_1} = (A_{\omega_1})^c$ and $(A^c)_{\omega_2} = (A_{\omega_2})^c$

(2) If for an index set I we have $A_\alpha \subseteq \Omega_1 \times \Omega_2$ for all $\alpha \in I$ then

$$(\cup_{\alpha \in I} A_\alpha)_{\omega_1} = \cup_{\alpha \in I} (A_\alpha)_{\omega_1}$$

and

$$(\cap_{\alpha \in I} A_\alpha)_{\omega_1} = \cap_{\alpha \in I} (A_\alpha)_{\omega_1}$$

and similarly for ω_2.

Definition 5.8. Suppose we have a funtion X with domain $\Omega_1 \times \Omega_2$ and range equal to some set S. We definethe section of the function X as

$$X_{\omega_1}(\omega_2) = X(\omega_1, \omega_2) \quad \forall \omega_2 \in \Omega_2$$

so $X_{\omega_1} : \Omega_2 \to S$ and similarly we can define $X_{\omega_2} : \Omega_1 \to S$.

Proposition 5.5. *Some properties of function sections are*

(1) $(1_A)_{\omega_1} = 1_{A_{\omega_1}}$

(2) If $S = \mathbb{R}^k$ for some $k \geq 1$ then for $i = 1, 2$ we have $X_i : \Omega_1 \times \Omega_2 \to S$ then $(X_1 + X_2)_{\omega_1} = (X_1)_{\omega_1} + (X_2)_{\omega_1}$

(3) Supose that S is a metric space and $X_n : \Omega_1 \times \Omega_2 \to S$ and the limit of X_n exists then

$$(\lim_{n \to \infty} X_n)_{\omega_1} = \lim_{n \to \infty} (X_n)_{\omega_1}$$

Definition 5.9 (Rectangles and RECT)**.** A rectangle in $\Omega_1 \times \Omega_2$ is a subset of $\Omega_1 \times \Omega_2$ of the form $A_1 \times A_2$ where $A_i \subseteq \Omega_i$. We call A_i the sides of the rectangle, we say that the rectangle is empty if at least one side is empty.Suppose that $(\Omega_i, \mathcal{B}_i)$ are measurable spaces for $i = 1, 2$ then we say that a rectangle is measurable if $A_i \in \mathcal{B}_i$ for $i = 1, 2$. We define the class of measurable rectangles as

$$\text{RECT} = \{A_1 \times A_2 : A_i \in \mathcal{B}_i\}$$

Remark: RECT is a semi-algebra.

Definition 5.10. We define a $\sigma-$field on $\Omega_1 \times \Omega_2$ generated by RECT as

$$\mathcal{B}_1 \otimes \mathcal{B}_2 = \sigma(\text{RECT})$$

We call $\mathcal{B}_1 \otimes \mathcal{B}_2$ the product $\sigma-$field. Note that if $\Omega_1 = \Omega_2 = \mathbb{R}$ this defines the Borel $\sigma-$field on $\mathbb{R}^2$.

Lemma 5.1. *Sections of measurable sets are measurable. That is if $A \in \mathcal{B}_1 \otimes \mathcal{B}_2$ then $A_{\omega_1} \in \mathcal{B}_2$ and $A_{\omega_2} \in \mathcal{B}_1$ for all $\omega_i \in \Omega_i$.*

Proof. We proceed by set induction. Let

$$\mathcal{D} = \{A \in \mathcal{B}_1 \otimes \mathcal{B}_2 : A_{\omega_1} \in \mathcal{B}_2,\ A_{\omega_2} \in \mathcal{B}_1\}.$$

Then RECT $\subseteq \mathcal{D}$ since for every $A = B_1 \times B_2 \in$ RECT, we have $A_{\omega_1} \in \{\emptyset, B_2\} \subseteq \mathcal{B}_2$ and $A_{\omega_2} \in \{\emptyset, B_1\} \subseteq \mathcal{B}_1$.

Moreover, $\mathcal{D}$ is a σ-algebra. Indeed, $\Omega_1 \times \Omega_2 \in \mathcal{D}$, and if $A \in \mathcal{D}$, then

$$(A^c)_{\omega_1} = (A_{\omega_1})^c, \qquad (A^c)_{\omega_2} = (A_{\omega_2})^c,$$

so $(A^c)_{\omega_1} \in \mathcal{B}_2$ and $(A^c)_{\omega_2} \in \mathcal{B}_1$. Also, if $A_n \in \mathcal{D}$, then

$$\left(\bigcup_n A_n\right)_{\omega_1} = \bigcup_n (A_n)_{\omega_1} \in \mathcal{B}_2$$

and similarly $\left(\bigcup_n A_n\right)_{\omega_2} \in \mathcal{B}_1$.

Therefore, by the σ-induction principle, $\mathcal{B}_1 \otimes \mathcal{B}_2 \subseteq \mathcal{D}$. Since $\mathcal{D} \subseteq \mathcal{B}_1 \otimes \mathcal{B}_2$ by definition, we conclude that $\mathcal{D} = \mathcal{B}_1 \otimes \mathcal{B}_2$. □

Corollary 5.3. *Sections of measurable functions are measurable. That is if $X : \Omega_1 \times \Omega_2 \to S$ is $\mathcal{B}_1 \otimes \mathcal{B}_2/\mathcal{B}(S)$ measurable then $X_{\omega_1} : \Omega_2 \to S$ and $X_{\omega_2} : \Omega_1 \to S$ are $\mathcal{B}_2/\mathcal{B}(S)$ and $\mathcal{B}_1/\mathcal{B}(S)$ measurable respectively.*

Proof. Let $B \in \mathcal{B}(S)$ then $X^{-1}(B) \in \mathcal{B}_1 \otimes \mathcal{B}_2$ and therefore $(X^{-1}(B))_{\omega_1} \in \mathcal{B}_2$ and $(X^{-1}(B))_{\omega_2} \in \mathcal{B}_1$. Now note that

$$(X^{-1}(B))_{\omega_1} = \{\omega_2 : X(\omega_1, \omega_2) \in B\} = X_{\omega_1}^{-1}(B),$$

and similarly $(X^{-1}(B))_{\omega_2} = X_{\omega_2}^{-1}(B)$. Therefore, X_{ω_1} is $\mathcal{B}_2/\mathcal{B}(S)$ measurable and X_{ω_2} is $\mathcal{B}_1/\mathcal{B}(S)$ measurable. □

5.8 Probabiltiy Measures on Product Spaces

Definition 5.11 (Transition Functions)**.** We call a function

$$K(\omega_1, A_2) : \Omega_1 \times \mathcal{B}_2 \to [0, 1]$$

a transtition function if

(1) For each $\omega_1 \in \Omega_1$, $K(\omega_1, \cdot)$ is a probability measure on $\mathcal{B}_2$.

(2) For each $A_2 \in \mathcal{B}_2$, $K(\cdot, A_2)$ is $\mathcal{B}_1/\mathcal{B}([0, 1])$ measurable.

Theorem 5.6. *Let P_1 be a probability measure then $K and P_1$ uniquely determine a probability on $\mathcal{B}_1 \times \mathcal{B}_2$ by*

$$P(A_1 \times A_2) = \int_{A_1} K(\omega_1, A_2) dP_1(\omega_1)$$

for all $A_1 \times A_2 \in RECT$ and this extends uniquely to $\mathcal{B}_1 \times \mathcal{B}_2$.

5.9 Fubini's Theorem

Theorem 5.7. *Let P_1 be a probability measure on $(\Omega_1, \mathcal{B}_1)$ and let K be a transition function. Let P be the probability measure on $\mathcal{B}_1 \times \mathcal{B}_2$ defined by*

$$P(A_1 \times A_2) = \int_{A_1} K(\omega_1, A_2) P_1(d\omega_1)$$

Assume that $X : (\Omega_1 \times \Omega_2, \mathcal{B}_1 \times \mathcal{B}_2) \to (\mathbb{R}, \mathcal{B}(\mathbb{R}))$ and furthermore suppose $X \geq 0$ then

$$Y(\omega_1) = \int_{\Omega_2} K(\omega_1, d\omega_2) X_{\omega_1}(\omega_2)$$

It has the following properties:

(1) Y is well defined

(2) $Y \in \mathcal{B}_1$

(3) $Y \geq 0, Y \in \ell_1(P_1)$

Furthermore we have that

$$\int_{\Omega_1 \times \Omega_2} X dP = \int_{\Omega_1} Y dP_1 = \int_{\Omega_1} \int_{\Omega_2} K(\omega_1, d\omega_2) X_{\omega_1}(\omega_2) P_1(d\omega_1)$$

Chapter 6

Convergence

6.1 Almost Sure Convergence

Definition 6.1. A sequence of random variables $\{X_n\}$ converges almost surely to a random variable X if for every $\varepsilon > 0$ and every $\delta > 0$, there exists $N \in \mathbb{N}$ such that

$$P\left(|X_n(\omega) - X(\omega)| < \varepsilon \text{ for all } n \geq N\right) > \delta.$$

We write this as $X_n \to X$ a.s.

6.2 Convergence in Probability

Definition 6.2. Suppose X_n is a sequence of random variables and X is a random variable. We say that X_n converges in probability to X if for all $\varepsilon > 0$ we have

$$\lim_{n\to\infty} P(|X_n - X| > \varepsilon) = 0$$

We write this as $X_n \to X$ i.p.

Theorem 6.1. *Convergence almost surely implies convergence in probability. That is if $X_n \to X$ a.s then $X_n \to X$ in probability.*

Proof. If $X_n \to X$ a.s then for all $\varepsilon > 0$ we have that $P(|X_n - X| > \varepsilon \text{ infinitely often}) = 0$. Therefore, for all $\varepsilon > 0$ we have that $P(\limsup |X_n - X| > \varepsilon) = 0$ so $\lim_{N\to\infty} P(\cup_{n\geq N} |X_n - X| > \varepsilon) = 0$ and therefore $\lim_{N\to\infty} \sup_{n\geq N} P(|X_n - X| > \varepsilon) = 0$ which implies that $\lim_{n\to\infty} P(|X_n - X| > \varepsilon) = 0$. □

Theorem 6.2. *Suppose that X_n is a sequence of real valued random variables*

(1) Cauchy Criterion: X_n converges i.p if and only if X_n is Cauchy i.p. That is for all $\varepsilon > 0$ we have

$$\lim_{n\to\infty} \sup_{m\geq n} P(|X_n - X_m| > \varepsilon) = 0$$

(2) $X_n \to X$ i.p if and only if each subsequence of $\{X_n\}$ has a further subsequence that converges to X a.s.